

Generator

If we have a function with eigenvalues given by

$$\hat{H}|E\rangle = E|E\rangle \quad (1)$$

And a generator using that

$$\hat{U}(t)|E\rangle = e^{\frac{-i}{\hbar}\hat{H}t}|E\rangle \quad (2)$$

$$= e^{\frac{-i}{\hbar}Et}|E\rangle \quad (3)$$

We just throw in the eigenvalue because the $\hat{H}$ always applies to $|E\rangle$ in the Taylor expansion.